# Deploying NETRAX on existing CCTV

Lecture notes for the jury · Team ZENTRIX · BH26PS06T044 · 12 Nov 2026

**The answer in one sentence**

NETRAX does not need its own cameras: it reads the live stream that every modern CCTV camera or recorder already provides (RTSP) over the site network, runs the AI on our edge computer, and leaves the CCTV recording exactly as it is.

## Part 1 · The idea (about 1 minute)

“Respected jury, most factories already have CCTV cameras and a recorder in the security room. Those cameras record, but nobody watches every screen all day. NETRAX turns that same CCTV into a safety officer.

Every IP CCTV camera, and every NVR or DVR recorder, can share its live picture on the local network through a standard called **RTSP** (Real-Time Streaming Protocol). It is the same stream the guard’s monitor or the mobile app uses. NETRAX simply becomes one more viewer of that stream. We do not replace cameras, we do not rewire anything and we do not touch the recording.”

## Part 2 · What connects to what (about 1 minute)

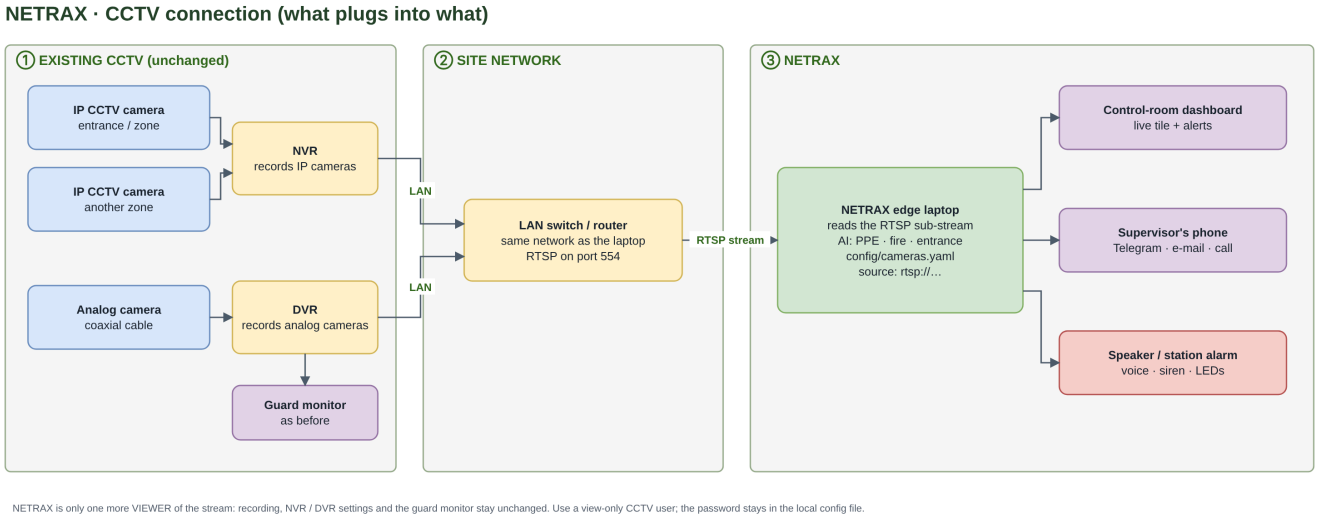



Figure 1 · CCTV connection diagram. Left: the existing CCTV, unchanged. Middle: the site network. Right: NETRAX reading the stream and raising alerts.

“On the left is the existing CCTV. IP cameras go to the NVR; old analog cameras go to a DVR by coaxial cable. Both recorders are already on the site network. Our edge laptop joins the same network and reads the stream on port 554. The AI checks PPE, fire and smoke, or does the entrance check, and the result appears on our dashboard, on the supervisor’s phone and on the alarm, exactly like with our own cameras.”

# Part 3 · Step by step (about 2 minutes)

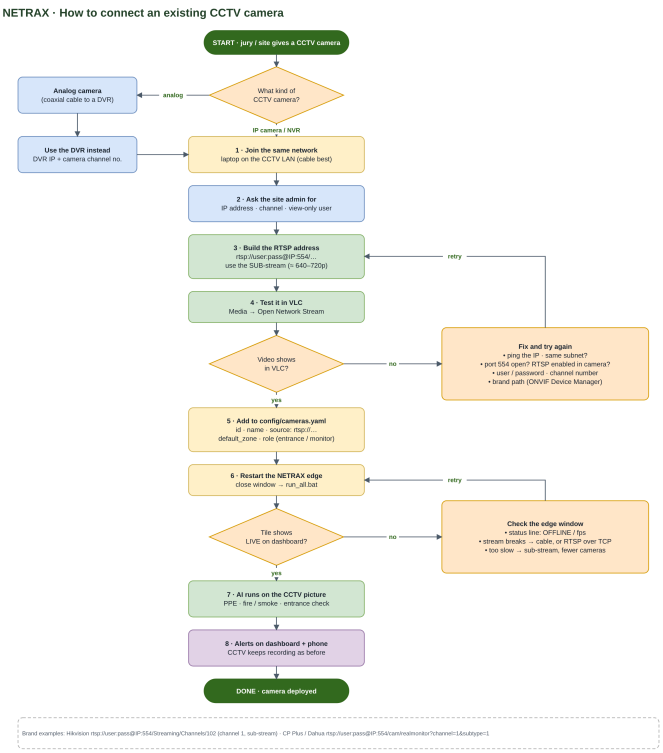



Figure 2 · Deployment flowchart.

**Step 0 · Which camera?** An IP camera or NVR gives a stream directly. An analog camera has no network, so we use the DVR it is connected to and choose that camera's channel number.

**Step 1 · Same network.** The NETRAX laptop joins the CCTV network, preferably with a cable.

**Step 2 · Ask the site admin** for the camera or recorder IP address, the channel number and a **view-only** user, so NETRAX can watch but never change the CCTV settings.

**Step 3 · Build the RTSP address.** Each brand has a fixed pattern (box below). We use the **sub-stream** (about 640–720p): our AI works at 640 pixels anyway, so nothing is lost and the network load is small.

**Step 4 · Test in VLC.** If VLC plays the address, NETRAX will too. If not: check the IP (ping), port 554, that RTSP is enabled, the user and password, the channel number.

**Step 5 · One entry in the settings file** `config/cameras.yaml`: name, the RTSP address, the zone, and the role (entrance or monitoring). No code changes.

**Step 6 · Restart the edge.** The camera appears as a live tile on the control room.

**Step 7–8 · Working.** The AI checks every person, alerts reach the dashboard and phone, and the CCTV keeps recording as before.

## RTSP address patterns (examples)

| Brand | Address (channel 1, sub-stream) |
|---|---|
| Hikvision | `rtsp://user:pass@192.168.1.64:554/Streaming/Channels/102` (channel 3 → `302`) |
| CP Plus / Dahua | `rtsp://user:pass@192.168.1.108:554/cam/realmonitor?channel=1&subtype=1` |
| Any other brand | From the camera manual, or found with the free tool "ONVIF Device Manager". |

## The only change in NETRAX: one block in `config/cameras.yaml`

```
cam_04:
  name: Main entrance
  source: rtsp://viewer:password@192.168.1.64:554/Streaming/Channels/102
  default_zone: main_entrance
  role: entrance          # or leave out for a normal monitoring camera
```

## Part 4 · Questions the jury may ask

| Question | Answer |
|---|---|
| Does it disturb the existing CCTV? | No. Reading a stream is what the guard monitor and mobile app already do. Recording and settings stay unchanged; we use a view-only user and the light sub-stream. |
| Is it supported in your code today? | Yes: the camera reader (`edge/stream.py`) accepts `rtsp://` addresses, and its health check uses port 554. Honestly, we have tested phones and USB cameras live, but not yet a real site CCTV. |
| What about old analog cameras? | We read them through the DVR’s RTSP stream, one channel per camera. |
| How many CCTV cameras? | One edge laptop (GTX 1650) handled 4 cameras live in our measurements. More cameras need another edge box; each box reads its own group of channels. |
| Is the video sent to the cloud? | No. The stream stays on the site network; only small alerts with face-blurred snapshots are stored. The CCTV password stays in a local file that is never uploaded. |
| Does placement matter? | Yes. Overhead CCTV sees helmets well but not always hands or feet; then gloves and shoes are “cannot tell”, never a false alarm. For the entrance, chest height at 2–4 m, one person at a time, works best. |
| What if the stream breaks on Wi-Fi? | Use a cable. NETRAX reconnects by itself; for unstable networks RTSP can be forced over TCP with one setting in the start script. |

### Closing line

“So NETRAX upgrades the CCTV a factory already owns. Plug our edge box into the same network, add one line per camera, and every existing camera starts checking PPE and fire, without new cameras and without touching the recording.”